

PRACTICE EXAM I
PHYS 5103 – Classical Mechanics
75 minutes
September 24, 2026

STUDENT _____

This exam consists of 2 derivations and 1 problem

Derivation 1

Suppose that a system of two particles obeys the following equations of motion (where $\vec{R}$ is the position vector of the center of mass, and $M=m_1+m_2$ is the total mass of the system)

$$M \frac{d^2 \vec{R}}{dt^2} = \sum_i \vec{F}_i^{(e)} = \vec{F}^{(e)}$$

and

$$\frac{d\vec{L}}{dt} = \sum_i \vec{N}_i^{(e)} = \vec{N}^{(e)}$$

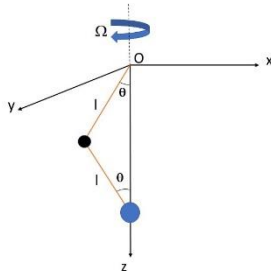
Using the equations of motion of the individual particles show that the internal forces between the particles obey (a) the weak law of action and reaction, and (b) the strong law of action and reaction.

Derivation 2

- (a) For circular and parabolic orbits in an attractive $1/r$ potential having the same angular momentum, show that the perihelion distance of the parabola is one-half the radius of the circle.
- (b) Prove that in the same central force as in part (a) the speed of a particle at any point in a parabolic orbit is $\sqrt{2}$ times the speed in a circular orbit passing through the same point.

PROBLEM 1

Consider a mass m attached with massless rods of length l to a fixed point at the origin, and to a bead of mass m on the z -axis, as shown in the figure. The whole system is made to rotate with constant angular velocity Ω about the vertical axis. The bead can move up or down on the axis. The gravitational force is assumed uniform, pointing down in the z direction. The angle the top rod makes with vertical axis is θ , which we choose as the single generalized coordinate needed for the system.



- (a) Write Lagrange's equation for θ .
- (b) How many equilibrium configurations are there? Describe the equilibrium positions in the limits for slow and fast rotation (small and large Ω).